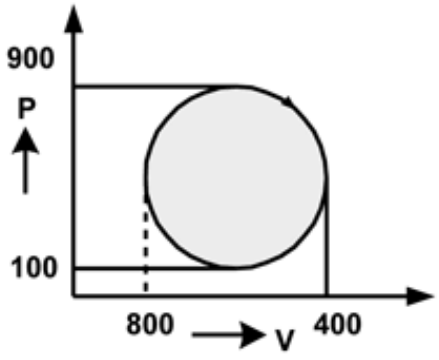
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

How much heat is absorbed by the system in going through the process shown in the figure.
(Consider that value is taken in SI system)



(A) 20.4×10^4 J

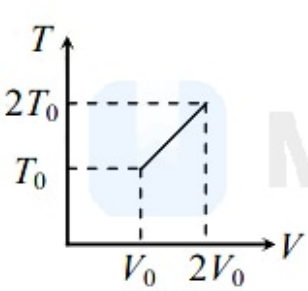
(B) 30.4×10^4 J

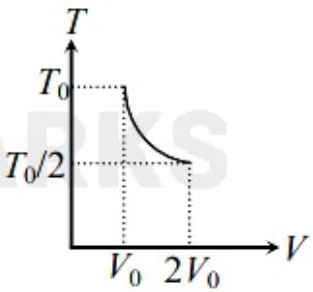
(C) 21.4×10^4 J

(D) 25.12×10^4 J

Q.2

For two thermodynamic process temperature and volume, diagrams are drawn. In the first process, it is a straight line having initial and final coordinates as V_0, T_0 and $2V_0, \quad 2T_0$, whereas in second process it is a rectangular hyperbola having initial and final coordinates (V_0, T_0) and ($2V_0$, $T_0/2$). Then the ratio of work done in the two processes must be





(A) 1 : 2

(B) 2 : 1

(C) 1 : 1

(D) None of these

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Q.3

When 20 J of work was done on a gas, 40 J of heat energy was released. If the initial internal energy of the gas was 70 J, what is the final internal energy?

(A) 50 J

(B) 60 J

(C) 90 J

(D) 110 J

Q.4

How much mechanical work must be done to completely melt 1 gram of ice at 0°C?

(A) 4.2 J

(B) 80 J

(C) 336 J

(D) 2268 J

Q.5

Which of the following is incorrect regarding the first law of thermodynamics?

(A) It is applicable to cyclic process.

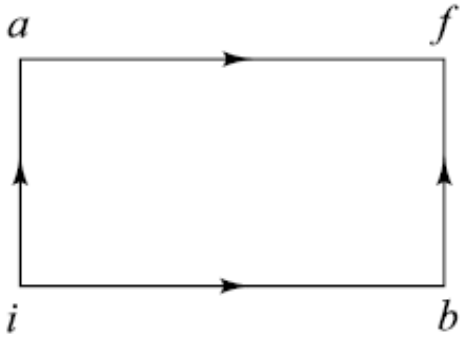
(B) It is a restatement of the principle of conservation of energy.

(C) It introduces the concept of internal energy.

(D) It introduces the concept of entropy.

Q.6

When a system is taken from state i to state f along the path iaf, it is found that $Q = 50$ cal and $W = 20$ cal. Along the path ibf, $Q = 36$ cal. W along the path ibf is [see the figure]



(A) 6cal

(B) 16cal

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(C) 66cal

(D) 14cal

Q.7

The work of $146\ \text{kJ}$ is performed in order to compress one kilo mole of gas adiabatically and in this process the temperature of the gas increases by 7°C. The gas is a $\left(R = 8.3\ \text{J mol}^{-1}\ \text{K}^{-1}\right)$

(A) mixture of monoatomic and diatomic gases

(B) monoatomic gas

(C) diatomic gas

(D) triatomic gas

Q.8

Two rigid boxes containing different ideal gases are placed on a table. $\text{Box}\ A$ contains on mole of nitrogen at temperature T_0, while box B contains one mole of helium at temperature $(7/3)T_0$. The boxes are then put into thermal contact with each other, and heat flows between them until the gases reach a common final temperature. (Ignore the heat capacity of boxes). Then the final temperature of the gases, T_f, in terms of T_0 is

(A) $T_f = \frac{3}{2}T_o$

(B) $T_f = \frac{5}{2}T_o$

(C) $T_f = \frac{3}{7}T_0$

(D) $T_f = \frac{7}{3}T_0$

Q.9

Which of the following statements is correct for any thermodynamic system?

(A) The internal energy changes in all processes.

(B) Internal energy and entropy are state functions.

(C) The change in entropy can never be zero.

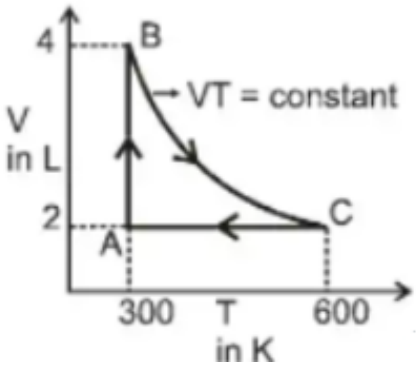
(D) The work done in an adiabatic process is always zero.

Q.10

Two moles of an ideal monoatomic gas undergoes a cyclic process ABCA as shown in the $\text{V} - \text{T}$ diagram above.
Then, which of the following statement is correct?

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[R is gas constant]



(A) Work done on the gas during the BC process is $500\mathrm{R}$

(B) Heat supplied to the gas during the BC process is $1500\mathrm{R}$

(C) Heat supplied to the gas during the AB process is $460\mathrm{R}$ In2

(D) Work done by the gas during the entire cycle is $600\mathrm{R}(1-\ln 2)$

Q.11

One mole of an ideal gas with adiabatic exponent γ whose pressure changes with volume as $P=\alpha V$, where α is a constant, is expanded so that its volume increase η times. Find the change in internal energy and heat capacity of of gas is V_0.

(A) $(\gamma-1)\left[1-\eta^2\right]/\alpha v_0^2, C=Rx/2$

(B) $\alpha v_0^2\left[\eta^2-1\right]/(\gamma-1), C=R[\gamma+1]/2[\gamma-1]$

(C) $v_0^2\gamma\left[1-\eta^2\right]/(\alpha-1), C=R[\gamma+1]/2[\gamma-1]$

(D) $\alpha v_0^2\left[\eta^2-1\right]/(\gamma-1), C=R[\gamma-1]/2[\gamma+1]$

Q.12

One mole of a monoatomic ideal gas is contained in a insulated and rigid container. It is heated by passing a current of 2 A for 10 minutes through a filament of resistance 100Ω. The change in the internal energy of the gas is

(A) $30\ \mathrm{kJ}$

(B) $60\ \mathrm{kJ}$

(C) $120\ \mathrm{kJ}$

(D) $240\ \mathrm{kJ}$

Q.13

$743\ \mathrm{J}$ of heat energy is needed to raise the temperature of 5 moles of an ideal gas by $2\ \mathrm{K}$ at constant pressure. How much heat energy is needed to raise the temperature of the same mass of the gas by $2\ \mathrm{K}$ at constant volume? Given, molar gas constant $R=8.3\ \mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$.

(A) $826\ \mathrm{J}$

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(B) $743\ \text{J}$

(C) $660\ \text{J}$

(D) none of these

Q.14

An amount Q of heat is added to a monatomic ideal gas in a process in which the gas perfomrs a work Q/2 on its surrounding.Find the molar heat capacity for the process.

(A) 3R

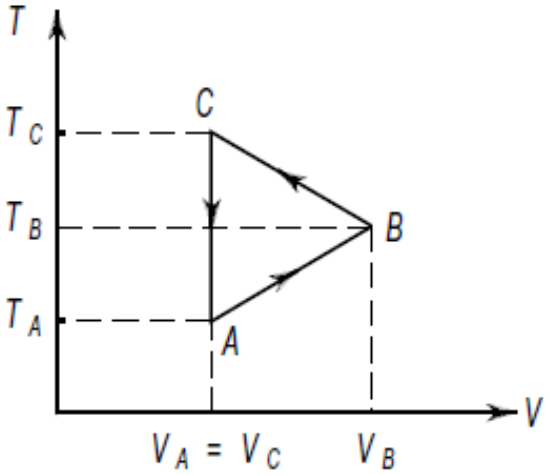
(B) 4R

(C) 6R

(D) 7R

Q.15

Three moles of an ideal gas are taken through a cyclic process $ABCA$ as shown on $T-V$ diagram in Fig. 8.24. The gas loses $2510\ \text{J}$ of heat in the complete cycle. If $T_A = 100\ \text{K}$ and $T_B = 200\ \text{K}$, The work done by the gas during the process BC is



(A) $5000\ \text{J}$

(B) $-5000\ \text{J}$

(C) $4000\ \text{J}$

(D) $-2500\ \text{J}$

Q.16

A body initially at 80°C cools to 64°C in 5 minutes and to 52°C in 10 minutes. The temperature of the surroundings is

(A) 15°C

(B) 16°C

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(C) $20^{\circ}\mathrm{C}$

(D) $25^{\circ}\mathrm{C}$

Q.17

An ideal gas $(\gamma = 1.4)$ expands from $5 \times 10^{-3}\ \mathrm{m}^3$ to $25 \times 10^{-3}\ \mathrm{m}^3$ at a constant pressure of $1 \times 10^5\ \mathrm{Pa}$. The heat energy supplied to the gas in this process is

(A) 7 J

(B) $70\ \mathrm{J}$

(C) $700\ \mathrm{J}$

(D) $7000\ \mathrm{J}$

Q.18

A monoatomic ideal gas, initially at temperature T_1, is enclosed in a cylinder fitted with a frictionless piston. The gas is allowed to expand adiabatically to a temperature T_2 by releasing the piston suddenly. If L_1 and L_2 are the lengths of the gas column before and after expansion respectively, then T_1/T_2 is given by

(A) $\left(\frac{L_1}{L_2}\right)^{2/3}$

(B) $\frac{L_1}{L_2}$

(C) $\frac{L_2}{L_1}$

(D) $\left(\frac{L_2}{L_1}\right)^{2/3}$

Q.19

5 moles of an ideal gas are compressed to half the initial volume at a constant temperature of $27.0^{\circ}\mathrm{C}$. Calculate the work done in the process. Given $R = 8.3\ \mathrm{J\ K^{-1}\ mol^{-1}}$. Write your result up to appropriate significant figures.

(A) W = – 8.6 $\times 10^3$ J

(B) W = – 7.6 $\times 10^3$ J

(C) W = – 6.6 $\times 10^3$ J

(D) 15W = – 9.6 $\times 10^3$ J

Q.20

A diatomic gas does $100\ \mathrm{J}$ of work when expanded at constant pressure. The heat energy absorbed by the gas in this process is

(A) $200\ \mathrm{J}$

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(B) 250 J

(C) 300 J

(D) 350 J

Answers & Solutions

Q.1 Answer:

25.12×10^4 J

Solution:

Work done = Area of PV diagram

$= \frac{\pi}{4}(P_2 - P_1)(V_2 - V_1)$

$= \frac{\pi}{4}(800)400$

$= 800 \times 100 \times 3.14$

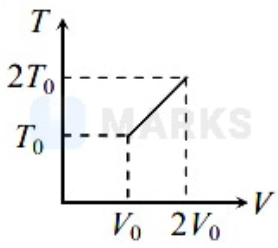
$= 8 \times 10^4 \times 3.14$

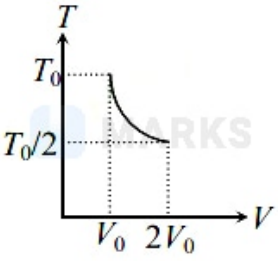
$= 25.12 \times 10^4$

Q.2 Answer:

2 : 1

Solution:





The first process is the constant pressure process Hence, $w_1 = nR(2T_0 - T_0) = nRT_0$

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Equation of the second process is $T = \frac{c}{V}$

Hence, $P = \frac{nRT}{V} = \frac{nRc}{V^2}$

$\therefore \quad w = \int_{V_0}^{2V_0} PdV = \frac{nRT_0}{2}$

$\therefore \quad \frac{w_1}{w_2} = 2 : 1$

Q.3 Answer:

50 J

Solution:

From the first law of thermodynamics, $dQ = dU + dW$ **or** $dU = dQ - dW = 40 - 20 = 20$ J. **Since heat energy is released by the gas, its internal energy is decreased from** 70 J **to** $70 - 20 = 50$ J. **Hence the correct choice is (a).**

Q.4 Answer:

336 J

Solution:

Work done $=$ **latent heat of fusion of ice** $= 80$ **calories** $= 80 \times 4.2 = 336$ J. **Hence the correct choice is (c).**

Q.5 Answer:

It introduces the concept of entropy.

Solution:

The only incorrect statement is (d). The concept of entropy is related to the second law of thermodynamics.

Q.6 Answer:

6cal

Solution:

From the first law of thermodynamics, we have

$Q = W + \Delta U$

Where Q $=$ **heat energy supplied,** W $=$ **work done and** ΔU $=$ **change in internal energy. For the paths iaf and irm, we have**

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$50 = 20 + \Delta U$

$36 = \mathrm{W} + \Delta U$

and Eliminating ΔU from Eqs. (1) and (2) we get $W = 6$ cal.

Q.7 Answer:

diatomic gas

Solution:

An adiabatic process, the work is given by

$W = \frac{nR(T_1 - T_2)}{\gamma - 1}$

or $146 \times 10^3 = \frac{1000 \times 8.3 \times 7}{(\gamma - 1)} \quad (\because \Delta T = 7^\circ\mathrm{C} = 7\ \mathrm{K})$

which gives $\gamma = 1 + 0.4 = 1.4$. **Hence the gas is diatomic**

Q.8 Answer:

$T_f = \frac{3}{2} T_o$

Solution:

The change in internal energy of an ideal gas is given by

$\Delta U = nC_v \Delta T = nC_v \left(T_f - T_\mathrm{i}\right)$

where $n =$ number of moles and $T_i =$ initial temperature.

For a diatomic gas (such as nitrogen) $C_v = \frac{5}{2} R$ and for a monoatomic gas (such as helium) $C_v = \frac{3}{2} R$.

$\therefore \Delta U$ for oxygen $= 1 \times \frac{5R}{2} \times (T_f - T_o)$

and ΔU for helium $= 1 \times \frac{3R}{2} \left(T_f - \frac{7T_o}{3}\right)$

Since no heat is given to or taken from the surrounding, the total internal energy of the system cannot change. Therefore

$1 \times \frac{5R}{2} \times (T_f - T_o) + 1 \times \frac{3R}{2} \left(T_f - \frac{7T_o}{3}\right) = 0$

Which gives $T_f = \frac{3}{2} T_o$, which is choice (a).

Q.9 Answer:

Internal energy and entropy are state functions.

Solution:

The correct choice is (b). The internal energy does not change in an isothermal process. Entropy can change in an adiabatic process. Work done in an adiabatic process is not zero.

Q.10 Answer:

Work done by the gas during the entire cycle is $600\mathrm{R}(1 - \ln 2)$

Solution:

(A) Work done on the gas during the BC process is $\mathrm{w} = \mathrm{P}\Delta\mathrm{V} = \mathrm{nR}\Delta\mathrm{T} = 2 \times \mathrm{R} \times (600 - 300) = 600\mathrm{R}$

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Thus, the option A is Incorrect

(C) The process AB is isothermal.

Heat supplied to the gas during the AB **process is** $\mathrm{q}=-\mathrm{w}=\mathrm{nRT} \ln \frac{\mathrm{V}_2}{\mathrm{V}_1}=2 \mathrm{R} \times 300 \ln \frac{4}{2}=600 \mathrm{R} \ln 2$

Hence, the option C **is Incorrect**

(D) The work done during the process AC **is o since the volume remains constant.**

Work done by the gas during the entire cycle is $\mathrm{w}=600 \mathrm{R}-600 \mathrm{Rin} 2=600 \mathrm{R}(1-\ln 2)$

Thus, the option D **is correct.**

Q.11 Answer:

$\alpha v_0^2\left[\eta^2-1\right] /(\gamma-1), C=R[\gamma+1] / 2[\gamma-1]$

Solution:

Given process: $P=\alpha V$ **...(i)**

Reaaranging above equation we can write

$\mathrm{PV}^{-1}=\alpha(a$ **constant) ...(ii)**

Equation (ii) shows a polytropic process with the value of polytropic constant $N=-1$**. Thus, the molar heat capacity of gas on this process can be given as**

$C=\frac{R}{\gamma-1}+\frac{R}{1-N}=\frac{R}{\gamma-1}+\frac{R}{2}=\frac{R}{2}\left[\frac{\gamma+1}{\gamma-1}\right]$

Let during the process, temperature of gas changes from T_1 **to** T_2 **then change in internal energy of gas:**

$\Delta U=C_V \Delta T=\frac{R}{\gamma-1}\left(T_2-T_1\right)$ **...(iii)**

As $n=1$ **mole and** $C_V=\frac{R}{\gamma-1}$

As initial and final volumes of the gas are V_0 **and** ηV_0**.**

Thus the respective pressures in initial and final states are: $P=\alpha V_0$ **and** $p=a \eta V_0$

From gas law we can find initial temperatures as

$T_1=\frac{P_1 V_1}{R}=\frac{\alpha V_0 \cdot V_0}{R}$ and $T_2=\frac{P_2 V_2}{R}=\frac{\alpha \eta V_0 \cdot \eta V_0}{R}$

Now from equation (iii) change in internal energy of gas is given as

$\Delta U=\frac{R}{\gamma-1}\left[\frac{\alpha \eta^2 V_0^2}{R}-\frac{\alpha V_0^2}{R}\right]=\frac{\alpha V_0^2}{\gamma-1}\left[\eta^2-1\right]$

Q.12 Answer:

$240 \mathrm{~kJ}$

Solution:

The heat energy supplied is

$$\begin{aligned}\Delta Q & =I^2 R t \\ & =(2)^2 \times 100 \times(10 \times 60) \\ & =240 \times 10^3 \mathrm{~J}=240 \mathrm{~kJ}\end{aligned}$$

Since $\Delta V=0$**; work done** $\Delta W=0$**. From first law of thermodynamics,** $\Delta U=\Delta Q=240 \mathrm{~kJ}$**. Thus, the correct choice is (d).**

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Q.13 Answer:

660 J

Solution:

Heat energy needed to raise the temperature of n moles of a gas through ΔT kelvin at constant pressure is

$Q = nC_p\Delta T = 743 \text{ J (given)}$

Heat energy needed to raise the temperature of n moles of a gas through ΔT kelvin at constant volume is

$$\begin{aligned} Q' &= nC_v\Delta T = n\left(C_p - R\right)\Delta T \left(\because C_p - C_v = R\right) \\ &= nC_p\Delta T - nR\Delta T \\ &= 743 - 5 \times 8.3 \times 2 \\ &= 743 - 83 = 660 \text{ J} \end{aligned}$$

Hence the correct choice is (c).

Q.14 Answer:

3R

Solution:

Given,

Amount of heat given to the gas $= \mathrm{Q}$

So, $\Delta \mathrm{Q} = \mathrm{Q}$

Work done by the gas, $\Delta \mathrm{W} = \frac{\mathrm{Q}}{2}$

From the first law of thermodynamics,

$$\begin{aligned} &\Delta \mathrm{Q} = \Delta \mathrm{W} + \Delta \mathrm{U} \\ &\Rightarrow \Delta \mathrm{U} = \mathrm{Q} - \frac{\mathrm{Q}}{2} = \frac{\mathrm{Q}}{2} \end{aligned}$$

For a monoatomic gas,

$$\begin{aligned} &\Delta \mathrm{U} = \frac{3}{2}\mathrm{nRdT} \\ &\Rightarrow \frac{\mathrm{Q}}{2} = \mathrm{ndT} \times \frac{3}{2} \\ &\Rightarrow \mathrm{Q} = 3\mathrm{nRdT} \end{aligned}$$

Again, for expansion at constant pressure,

$\mathrm{Q} = \mathrm{nC_p dT}$

where c_p is the molar heat capacity at constant pressure.

$$\begin{aligned} &\text{So, } 3 \text{ Rnd T} = \mathrm{nC_p dT} \\ &\Rightarrow \mathrm{C_p} = 3\mathrm{R} \end{aligned}$$

Q.15 Answer:

-5000 J

Solution:

In process AB, the volume V increases linearly with temperature T. Hence process AB is isobaric (constant pressure). Therefore, work done in this process is

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$$W_{AB} = P\Delta V = nR\Delta T \quad (\because PV = nRT)$$
$$= nR\,(T_B - T_A)$$
$$= 3 \times 8.3 \times (200 - 100) = 2490 \text{ J}$$

Process CA is isochoric (constant volume). Hence work done in this process $W_{CA} = 0$. Since the whole process $ABCA$ is cyclic, the change in internal energy in the complete cycle is zero, i.e. $\Delta U = 0$. Now, from the first law of thermodynamics, (Given $Q = -2510$ J)

$$Q = \Delta U + W = \Delta U + W_{AB} + W_{BC} + W_{CA}$$

or $\quad -2510 = 0 + 2490 + W_{BC} + 0$

or $\quad W_{BC} = -2510 - 2490 = -5000$ J

The negative sign shows that the work is done by the gas. Thus, the correct choice is (b).

Q.16 Answer:

16°C

Solution:

If T_0 is the temperature of the surroundings, then we have

$\log_e\left(\frac{80-T_0}{64-T_0}\right) = 5K$

and $\log_e\left(\frac{64-T_0}{52-T_0}\right) = 5K$

Equating Eqs (1) and (2), we get

$\frac{80-T_0}{64-T_0} = \frac{64-T_0}{52-T_0}$

which gives $T_0 = 16^\circ\text{C}$, which is choice (b).

Q.17 Answer:

7000 J

Solution:

Work done on the gas is

$$\Delta W = P\Delta V = P\,(V_f - V_i)$$
$$= 1 \times 10^5 \times (25 - 5) \times 10^{-3}$$
$$= 2000 \text{ J}$$

The internal energy is given by $U = \frac{PV}{(\gamma-1)} \therefore \quad U_i = \frac{PV_i}{(\gamma-1)}, U_f = \frac{PV_f}{(\gamma-1)}$.

Therefore, change in internal energy is

$$\Delta U = U_f - U_i = \frac{P}{(\gamma-1)}\,(V_f - V_i)$$
$$= \frac{1 \times 10^5 \times (25-5) \times 10^{-3}}{(1.4-1)} = 5000 \text{ J}$$

From the first law of thermodynamics, the heat energy supplied to the gas is

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$$\Delta Q = \Delta W + \Delta U = 2000 + 5000$$
$$= 7000 \text{ J}$$

which is choice (d).

Q.18 Answer:

$\left(\frac{L_2}{L_1}\right)^{2/3}$

Solution:

For adiabatic process, $T_1 V_1^{(\gamma-1)} = T_2 V_2^{(\gamma-1)}$. Thus

$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{(\gamma-1)}$

For a monoatomic gas, $\gamma = 5/3$. Also $V_2/V_1 = L_2/L_1$ Hence $\frac{T_1}{T_2} = \left(\frac{L_2}{L_1}\right)^{\left(\frac{5}{3}-1\right)} = \left(\frac{L_2}{L_1}\right)^{2/3}$,

which is choice (d).

Q.19 Answer:

W = – 8.6 $\times 10^3$ J

Solution:

Given $T = 27.0^\circ\text{C} = 300$ K. Since T is constant, the process is isothermal

$$\therefore W = nRT \log_e \left(\frac{V_2}{V_1}\right)$$
$$= 5 \times 8.3 \times 300 \times \log_e \left(\frac{1}{2}\right)$$
$$= 5 \times 8.3 \times 300 \times (-0.693)$$
$$= -8628 \text{ J}$$

Since the value of R has two significant figures $(s.f.)$, the value of W must be rounded off to $2s.f.$ as

$W = -8.6 \times 10^3$ J

The negative sign indicates that work is done on the gas.

Q.20 Answer:

350 J

Solution:

For a diatomic gas $C_v = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$. The work done in the isobaric process is

$$W = P(V_2 - V_1)$$
$$= nRT_2 - nRT_1$$
$$= nR(T_2 - T_1)$$
$$T_2 - T_1 = \frac{W}{nR}$$

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Heat absorbed is $\Delta Q = nC_p\left(T_2 - T_1\right)$

$= nC_p \times \frac{W}{nR}$

$= C_p \times \frac{W}{R}$

$= \frac{7R}{2} \times \frac{100}{R} = 350\ \text{J}$

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